

Lab03 SQL

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```
library(DBI)
library(RSQLite)

# Conecta/Cria o arquivo
con <- dbConnect(RSQLite::SQLite(), "me315.sqlite")
```

SQL

```
SELECT * FROM PRESERVE
```

Table 1: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0

Hands-on

1. Exiba os valores de STATE e FEE da tabela PRESERVE mas sem duplicados.

```
SELECT DISTINCT STATE, FEE FROM PRESERVE
```

Table 2: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0
MA	0

2. Exiba todos os valores de DNO na tabela EMPLOYEE. Não exiba valores duplicados.

```
SELECT DISTINCT DNO FROM EMPLOYEE
```

Table 3: 3 records

DNO
20
10
40

3. Execute as QUERYS a seguir. Examine as tabelas resultantes e faça as observações relevantes.

- a) `SELECT DNO, SALARY FROM EMPLOYEE` Apenas mostra todos as informações relacionadas as colunas escolhidas sem nenhuma order em específico e também repete alguns valores

```
SELECT DNO, SALARY FROM EMPLOYEE
```

Table 4: 6 records

DNO	SALARY
20	2000
10	2000

DNO	SALARY
20	3000
40	500
10	400
20	9000

- b) `SELECT DISTINCT DNO, SALARY FROM EMPLOYEE` Resulta na mesma tabela do item a, apesar do uso do “DISTINCT”, ambas as colunas não compartilham o mesmo DNO e SALARY simultaneamente

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
```

Table 5: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

- c) `SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY`
Nenhuma linha foi removida, porém o DNO foi ordenado em ordem crescente, tendo a ordem da coluna DNO como prioridade e a de SALARY apenas seguindo a ordem proposta por DNO

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY;
```

Table 6: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

4. Exiba todas as informações sobre qualquer reserva natural em Montana que seja menor que 1.000 acres.

```
SELECT PNO, PNAME, STATE, FEE, ACRES FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
```

Table 7: 2 records

PNO	PNAME	STATE	FEE	ACRES
3	DANCING PRAIRIE	MT	0	680
40	SOUTH FORK MADISON	MT	0	121

5. Exiba todas as informações sobre qualquer reserva natural cujo valor de ACRES esteja entre 1200 e 15000, inclusive.

```
SELECT PNO, PNAME, STATE, FEE, ACRES FROM PRESERVE
WHERE ACRES > 1200 AND ACRES < 15000
```

Table 8: 0 records

PNO	PNAME	STATE	FEE	ACRES
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6. Exiba todas as informações sobre qualquer reserva natural localizada em Montana que não possua taxa de entrada e seja maior que 10.000 acres.

```
SELECT PNO, PNAME, STATE, FEE, ACRES FROM PRESERVE
WHERE STATE = 'MT' AND FEE = 0 AND ACRES > 10000
```

Table 9: 1 records

PNO	PNAME	STATE	FEE	ACRES
2	PINE BUTTE SWAMP	MT	0	15000

7. Exiba todas as informações sobre as reservas naturais localizadas em Montana ou Massachusetts.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' OR STATE = 'MT'
```

Table 10: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

8. Exiba todas as informações sobre qualquer reserva natural localizada em Montana ou qualquer reserva que seja menor que 1.000 acres.

```
SELECT PNO, PNAME, STATE, ACRES, FEE FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
```

Table 11: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

9. Exiba o número e o nome de qualquer reserva natural que possua taxa de entrada que não seja igual a zero.

```
SELECT PNO, FEE, PNAME FROM PRESERVE
WHERE FEE > 0
```

Table 12: 3 records

PNO	FEE	PNAME
5	3	HASSAYAMPA RIVER
80	3	RAMSEY CANYON
6	3	PAPAGONIA-SONOITA CREEK

10. Exiba o número e o nome das reservas naturais que não possuem taxa de entrada de US\$ 3,00 e não possuem taxa de US\$ 10,00.

```
SELECT PNO, PNAME, FEE FROM PRESERVE
WHERE FEE = 3 AND FEE != 10
```

Table 13: 3 records

PNO	PNAME	FEE
5	HASSAYAMPA RIVER	3
80	RAMSEY CANYON	3
6	PAPAGONIA-SONOITA CREEK	3

11. Exiba todas as informações sobre qualquer reserva natural localizada no Arizona que não possua taxa de entrada, ou qualquer reserva que seja menor que 100 acres (independentemente de seus valores de STATE e FEE).

```
SELECT * FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100
```

Table 14: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

12. Exiba todas as informações sobre qualquer reserva natural que seja menor que 1.000 acres e tenha taxa de entrada de zero dólares ou esteja localizada no Arizona.

```
SELECT * FROM PRESERVE
WHERE ACRES < 1000 AND (FEE = 0 OR STATE = 'AZ')
```

Table 15: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0

13. Selecione todas as informações sobre qualquer reserva natural com taxa de entrada que não seja maior que zero, ou qualquer outra reserva, independentemente de sua taxa, que esteja localizada em Montana e seja maior que 1.000 acres.

```
SELECT * FROM PRESERVE
WHERE (FEE = 0) OR STATE = 'MT' AND ACRES > 1000
```

Table 16: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

14. Exiba todas as informações sobre todas as reservas naturais, exceto aquelas reservas de Montana sem taxa de entrada.

```
SELECT * FROM PRESERVE
WHERE NOT (STATE = 'MT' AND FEE = 0)
```

Table 17: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

15. As seguintes cláusulas WHERE são logicamente equivalentes? Sim, são equivalentes
 WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)
```

Table 18: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)
```


Table 19: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

16. As seguintes cláusulas WHERE são logicamente equivalentes? Sim, são logicamente equivalentes WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)
```

Table 20: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)
```

Table 21: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0

PNO	PNAME	STATE	ACRES	FEE
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

17. As seguintes cláusulas WHERE são logicamente equivalentes? Não são logicamente equivalentes WHERE NOT (ACRES < 50 AND STATE = 'MA')

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 AND STATE = 'MA')
```

Table 22: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

WHERE NOT ACRES < 50 AND NOT STATE = 'MA'

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 23: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

WHERE ACRES >= 50 OR STATE <> 'MA'

```
SELECT * FROM PRESERVE
WHERE ACRES >= 50 OR STATE <> 'MA'
```

Table 24: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

18. As seguintes cláusulas WHERE são logicamente equivalentes? Não são logicamente equivalentes WHERE NOT (ACRES < 50 OR STATE = 'MA')

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 OR STATE = 'MA')
```

Table 25: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0

PNO	PNAME	STATE	ACRES	FEE
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

WHERE NOT ACRES < 50 OR NOT STATE = 'MA'

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 OR NOT STATE = 'MA'
```

Table 26: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

WHERE NOT ACRES < 50 AND NOT STATE = 'MA'

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 27: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0

PNO	PNAME	STATE	ACRES	FEE
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-31 19:46:43 -03"
```